

SUBHAM DANGAR

MSc in Data Science and AI, RKMVERI, Belur

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GitHub LinkedIn Portfolio



SUMMARY

M.Sc. in Data Science and Artificial Intelligence student at Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI). Skilled in Machine Learning, Deep Reinforcement Learning, NLP, and distributed AI applications, with hands-on experience developing scalable semantic search systems, autonomous AI agents, and deploying production-ready ML solutions using Python and modern AI frameworks.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Data Science, 2025–Present
CGPA – 6.3 (till sem 1)

Bankura Christian College
BSc (Honours) in Mathematics, 2022–25
CGPA – 8.01

Higher Secondary (XII), WBCHSE, 2022
94.8%

Secondary (X), WBBSE, 2020
92.14%

SKILLS

- Languages:** Python, C, C++, SQL
- ML/DL:** Supervised Learning, Semantic Search, Model Evaluation
- Frameworks/Libraries:** NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, LangChain, LangGraph, Hugging Face Transformers, Sentence-Transformers, FAISS, Dask, FastAPI
- Tools:** SQLite, Git/GitHub, VS Code, Excel, Docker, Google Colab

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, LLM, Agentic AI,
- Probability and Statistics, Linear Algebra
- Distributed Computing, Data Structures and Algorithms
- Database Management System

ACHIEVEMENTS & SCHOLARSHIP

- INSPIRE Scholarship – Govt. of India (Merit-based)
- Qualified IIT JAM 2025 – Mathematics
- Qualified CUET (PG) 2025 – Mathematics
- Qualified CUET (PG) 2025 – Computer Science

CERTIFICATES

- NPTEL – Python Programming
- NPTEL – C++ Programming

PROJECTS

Crop Disease Detection using Remote Sensing Images
Python, PyTorch, CNN, OpenCV, NumPy Feb 2026 – Apr 2026
Developed a CNN-based system for automated crop disease detection from remote sensing images using image preprocessing, data augmentation, and deep learning. Trained and evaluated multiple CNN models using accuracy, precision, recall, F1-score, and confusion matrix to assess classification performance.

Repository-Aware Autonomous Coding Agent
Python, FastAPI, Ollama, LangChain, LangGraph, FAISS, Sentence-Transformers Ongoing
Built a fully local autonomous coding agent that understands software repositories and automates code generation, editing, testing, and debugging using semantic retrieval and local LLMs via Ollama. Integrated LangChain, LangGraph, and FAISS for repository-aware context retrieval, workflow orchestration, and iterative execution with safe SEARCH/REPLACE code modifications.

Distributed Educational Video Search System
Python, Sentence-Transformers, Dask, SQLite Mar 2026 – May 2026
Developed a multilingual semantic video retrieval system that returns the most relevant YouTube timestamps using sentence-transformer embeddings and cosine similarity. Implemented Dask-based parallel search, SQLite semantic caching, result ranking based on relevance and view count, and a web fallback mechanism to improve retrieval reliability.

Spam Email Detection System
Python, Scikit-learn, TF-IDF, Flask Sep 2025 – Nov 2025
Developed an end-to-end spam email classifier using text preprocessing, TF-IDF feature extraction, and machine learning algorithms including Naive Bayes, Logistic Regression, and Random Forest. Compared model performance using standard evaluation metrics and deployed the best-performing model as a Flask web application for real-time prediction.

ACADEMIC RESEARCH

A Study on Countability
B.Sc. Dissertation Feb 2025 – Jun 2025
Studied finite, countable, and uncountable sets through rigorous mathematical proofs, including Cantor’s diagonal argument and the countability of N , Z , and Q . Explored foundational concepts of set theory and cardinality in discrete mathematics.